TabuLLM: AI Benchmark Report

1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

Audit Indicator	Global Session Mean (± Std)
Synthetic Fidelity (JSD)	59.8 ± 1.25
Correlation Stability	79.1 ± 1.64
Distribution Consistency	71.7 ± 3.82
Entropy Preservation	30.3 ± 1.68
Mean Z Score Drift	0.3 ± 0.01
Categorical Fidelity	0.0 ± 0.00
Privacy Risk	0.0 ± 0.00
Data Density	100.0 ± 0.00
Coverage Score	95.6 ± 1.50
Feature Accuracy	51.8 ± 1.17
Novelty Score	0.8 ± 0.01

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Dataset Audit: chronic_kidney_disease.csv

1. Dataset Portrait

Attribute	Value
Rows	400
Columns	25
Numeric features	25
Categorical features	0
Missing cells (%)	10.1%

2. Generation Strategy & LLM Config [1]

Parameter	Configuration
Algorithm (*12)	gpt-4o-mini
Folds (*13)	10
Context % (*14)	0%
Generation % (*15)	100%
Join context (*16)	False

3. Statistical Significance & Data Drift

Feature	Orig Mean	Aug Mean	Shift	P-Value	Status
age	51.483	53.166	+1.682	3.3%	OK
bp	76.469	75.250	-1.219	1.6%	OK
sg	1.017	1.017	+0.000	0.0%	OK
al	1.017	0.047	-0.970	95.4%	Drift
su	0.450	0.000	-0.450	100.0%	Drift
rbc	0.190	0.000	-0.190	100.0%	Drift
pc	0.227	0.022	-0.205	90.4%	Drift
pcc	0.106	0.000	-0.106	100.0%	Drift
ba	0.056	0.000	-0.056	100.0%	Drift
bgr	148.037	116.169	-31.868	21.5%	Warn
bu	57.426	37.031	-20.395	35.5%	Drift
sc	3.072	1.691	-1.382	45.0%	Drift

4. Quality & Fidelity Metrics (Mean ± Std)

Metric Category	Aggregated Score
Synthetic Fidelity (JSD) (*1)	59.8 ± 1.25
Correlation Stability (*2)	79.1 ± 1.64
Distribution Consistency (*3)	71.7 ± 3.82
Entropy Preservation (*4)	30.3 ± 1.68
Mean Z Score Drift (*5)	0.3 ± 0.01
Categorical Fidelity (*6)	0.0 ± 0.00
Privacy Risk (*7)	0.0 ± 0.00

Data Density (*8)	100.0 ± 0.00
Coverage Score (*9)	95.6 ± 1.50
Feature Accuracy (*10)	51.8 ± 1.17
Novelty Score (*11)	0.8 ± 0.01

*[1] (*12) Algorithm: Contextual constraint or model parameter.*

*[2] (*13) Folds: Contextual constraint or model parameter.*

*[3] (*14) Context %: Contextual constraint or model parameter.*

*[4] (*15) Generation %: Contextual constraint or model parameter.*

*[5] (*16) Join context: Contextual constraint or model parameter.*

[6] Synthetic Fidelity (JSD): Overall distribution similarity.
[7] Correlation Stability: Multivariate relationships preserved.
[8] Distribution Consistency: Median feature similarity.
[9] Entropy Preservation: Information diversity maintained.
[10] Mean Z Score Drift: Standardized distribution shift.
[11] Categorical Fidelity: Category distribution similarity.
[12] Privacy Risk: Synthetic rows too close to originals.
[13] Data Density: Non-missing cells in augmented data.
[14] Coverage Score: Synthetic coverage of original space.
[15] Feature Accuracy: Mean feature similarity.
[16] Novelty Score: Distance ratio to nearest neighbors.

6. Downstream ML Utility: Comparative Analysis

Model: RF

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.99875	0.9125	0.9862500000000001	-0.012499999999999978	*
F1	0.9989010989010989	0.9270642173176379	0.9885851817545616	-0.010315917146537346	*

Model: LR

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.9100000000000001	0.7500000000000001	0.8800000000000001	-0.030000000000000006	
F1	0.92384446063128	0.8036903256970295	0.9042723637897009	-0.01957209684157928	

Model: KNN

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6125	0.625	0.7050000000000001	0.0925	
F1	0.5885198185573248	0.7387915246167118	0.7657783192103816	0.1772585006530568	

Model: NB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.9525	0.91875	0.9525	-1.1102230246251566e-17	
F1	0.9587663012688215	0.9346931733884201	0.9593235708944515	0.00055726962562998 59	

Model: XGB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.97875	0.8699999999999999	0.9825000000000002	0.00374999999999999 77	
F1	0.9805471229310466	0.8905237726882099	0.985296429146999	0.00474930621595247 55	

Model: DT

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Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.9537500000000001	0.79375	0.9237499999999998	-0.029999999999999992	
F1	0.9574646620033119	0.819176602063963	0.9328104433530704	-0.024654218650241633	

Model: GB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.9887500000000001	0.84375	0.975	-0.013749999999999974	
F1	0.9903119429590017	0.8592004737322461	0.9782861397191303	-0.012025803239871624	

Model: MLP

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6637500000000001	0.53	0.62625	-0.037500000000000006	
F1	0.6377190651635288	0.5724498900142143	0.6716436145600545	0.03392454939652564	

Model: AB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.9825000000000002	0.8675	0.975	-0.007499999999999995	
F1	0.9848619694663338	0.8839999963791556	0.9793144565392653	-0.005547512927068374	

Model: SVM

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6024999999999999	0.59625	0.6024999999999999	0	
F1	0.7507214882964476	0.7446829466013597	0.7507214882964476	0	

Appendix: Benchmark Data Trace

Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline

A. Baseline: Original Dataset (Full Record Sample)

#	age	bp	sg	al	su	rbc	pc	pcc	ba	bgr
1	48.0	80.0	1.02	1.0	0.0	nan	0.0	0.0	0.0	121.0
2	7.0	50.0	1.02	4.0	0.0	nan	0.0	0.0	0.0	nan
3	62.0	80.0	1.01	2.0	3.0	0.0	0.0	0.0	0.0	423.0
4	48.0	70.0	1.005	4.0	0.0	0.0	1.0	1.0	0.0	117.0
5	51.0	80.0	1.01	2.0	0.0	0.0	0.0	0.0	0.0	106.0
6	60.0	90.0	1.015	3.0	0.0	nan	nan	0.0	0.0	74.0
7	68.0	70.0	1.01	0.0	0.0	nan	0.0	0.0	0.0	100.0
8	24.0	nan	1.015	2.0	4.0	0.0	1.0	0.0	0.0	410.0
9	52.0	100.0	1.015	3.0	0.0	0.0	1.0	1.0	0.0	138.0
10	53.0	90.0	1.02	2.0	0.0	1.0	1.0	1.0	0.0	70.0
11	50.0	60.0	1.01	2.0	4.0	nan	1.0	1.0	0.0	490.0
12	63.0	70.0	1.01	3.0	0.0	1.0	1.0	1.0	0.0	380.0
13	68.0	70.0	1.015	3.0	1.0	nan	0.0	1.0	0.0	208.0
14	68.0	70.0	nan	nan	nan	nan	nan	0.0	0.0	98.0
15	68.0	80.0	1.01	3.0	2.0	0.0	1.0	1.0	1.0	157.0
16	40.0	80.0	1.015	3.0	0.0	nan	0.0	0.0	0.0	76.0
17	47.0	70.0	1.015	2.0	0.0	nan	0.0	0.0	0.0	99.0
18	47.0	80.0	nan	nan	nan	nan	nan	0.0	0.0	114.0
19	60.0	100.0	1.025	0.0	3.0	nan	0.0	0.0	0.0	263.0
20	62.0	60.0	1.015	1.0	0.0	nan	1.0	1.0	0.0	100.0

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B.1 Audit Trace: Fold 1 | Gen: 100%

#	age	bp	sg	al	su	rbc	pc	pcc	ba	bgr
1	65.0	70.0	1.0180976709267249	0.0	0.0	0.0	0.0	0.0	0.0	138.0
2	55.0	70.0	1.0155914521233584	0.0	0.0	0.0	0.0	0.0	0.0	92.0
3	54.0	80.0	1.0149528203243143	0.0	0.0	0.0	0.0	0.0	0.0	113.0
4	42.0	80.0	1.0158281285970063	0.0	0.0	0.0	0.0	0.0	0.0	120.0
5	62.0	80.0	1.013381620734987	0.0	0.0	0.0	1.0	0.0	0.0	112.0
6	57.0	70.0	1.0113050182444103	0.0	0.0	0.0	0.0	0.0	0.0	94.0
7	35.0	80.0	1.005	3.0	0.0	1.0	0.0	0.0	0.0	nan
8	30.0	80.0	1.025	0.0	0.0	0.0	0.0	0.0	0.0	96.0
9	52.0	80.0	1.025	0.0	0.0	0.0	0.0	0.0	0.0	99.0
10	65.0	80.0	1.015	0.0	0.0	nan	0.0	0.0	0.0	115.0

B.2 Audit Trace: Fold 2 | Gen: 100%

#	age	bp	sg	al	su	rbc	pc	pcc	ba	bgr
1	24.0	70.0	1.0171488957115627	0.0	0.0	0.0	0.0	0.0	0.0	102.0
2	59.0	70.0	1.014584297524655	0.0	0.0	0.0	0.0	0.0	0.0	141.0
3	60.0	70.0	1.0189272577333217	0.0	0.0	0.0	0.0	0.0	0.0	140.0
4	18.0	70.0	1.0195115363024403	0.0	0.0	0.0	0.0	0.0	0.0	119.0
5	14.0	70.0	1.0189338657205849	1.0	0.0	0.0	0.0	0.0	0.0	142.0
6	31.0	80.0	1.0173729068419222	0.0	0.0	0.0	0.0	0.0	0.0	119.0
7	64.0	70.0	nan	nan	nan	nan	nan	0.0	0.0	113.0
8	42.0	80.0	1.02	0.0	0.0	0.0	0.0	0.0	0.0	98.0
9	12.0	60.0	1.015	3.0	0.0	1.0	1.0	1.0	0.0	nan
10	23.0	80.0	1.025	0.0	0.0	0.0	0.0	0.0	0.0	111.0

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B.3 Audit Trace: Fold 3 | Gen: 100%

#	age	bp	sg	al	su	rbc	pc	pcc	ba	bgr
1	68.0	70.0	1.0182596472824004	0.0	0.0	0.0	0.0	0.0	0.0	123.0
2	38.0	80.0	1.0155521321903478	0.0	0.0	0.0	0.0	0.0	0.0	88.0
3	62.0	70.0	1.01806084062798	0.0	0.0	0.0	0.0	0.0	0.0	105.0
4	64.0	70.0	1.0142877461038107	0.0	0.0	0.0	0.0	0.0	0.0	106.0
5	16.0	80.0	1.0197087019870426	0.0	0.0	0.0	0.0	0.0	0.0	113.0
6	46.0	70.0	1.0173446953329732	0.0	0.0	0.0	0.0	0.0	0.0	122.0
7	34.0	80.0	1.025	0.0	0.0	0.0	0.0	0.0	0.0	121.0
8	33.0	60.0	nan	nan	nan	0.0	0.0	0.0	0.0	130.0
9	60.0	70.0	nan	nan	nan	nan	nan	0.0	0.0	124.0
10	57.0	60.0	1.02	0.0	0.0	0.0	0.0	0.0	0.0	132.0

B.4 Audit Trace: Fold 4 | Gen: 100%

#	age	bp	sg	al	su	rbc	pc	pcc	ba	bgr
1	66.0	70.0	1.0167928640558466	2.0	0.0	0.0	0.0	0.0	0.0	105.0
2	77.0	70.0	1.0184049466165495	0.0	0.0	0.0	0.0	0.0	0.0	96.0
3	68.0	70.0	1.0127965798934946	1.0	0.0	0.0	0.0	0.0	0.0	116.0
4	54.0	80.0	1.0210563398391321	0.0	0.0	0.0	0.0	0.0	0.0	100.0
5	62.0	80.0	1.0174772408392794	0.0	0.0	0.0	0.0	0.0	0.0	106.0
6	62.0	70.0	1.0184969272339672	0.0	0.0	0.0	0.0	0.0	0.0	122.0
7	44.0	60.0	1.02	0.0	0.0	0.0	0.0	0.0	0.0	95.0
8	55.0	90.0	1.01	2.0	1.0	1.0	1.0	0.0	0.0	273.0
9	45.0	90.0	nan	nan	nan	nan	nan	0.0	0.0	360.0
10	72.0	90.0	nan	nan	nan	nan	nan	0.0	0.0	308.0

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B.5 Audit Trace: Fold 5 | Gen: 100%

#	age	bp	sg	al	su	rbc	pc	pcc	ba	bgr
1	59.0	70.0	1.01886212352118 72	0.0	0.0	0.0	0.0	0.0	0.0	80.0
2	65.0	80.0	1.02065312850722 13	0.0	0.0	0.0	0.0	0.0	0.0	127.0
3	59.0	80.0	1.02064606574272	0.0	0.0	0.0	0.0	0.0	0.0	151.0
4	72.0	70.0	1.01601236903974 73	0.0	0.0	0.0	0.0	0.0	0.0	96.0
5	43.0	80.0	1.02088564734646 83	0.0	0.0	0.0	0.0	0.0	0.0	123.0
6	72.0	70.0	1.01698915335872 28	0.0	0.0	0.0	0.0	0.0	0.0	102.0
7	41.0	80.0	1.015	1.0	4.0	1.0	0.0	0.0	0.0	210.0
8	34.0	70.0	1.025	0.0	0.0	0.0	0.0	0.0	0.0	87.0
9	26.0	70.0	1.015	0.0	4.0	nan	0.0	0.0	0.0	250.0
10	50.0	80.0	1.02	0.0	0.0	0.0	0.0	0.0	0.0	137.0

B.6 Audit Trace: Fold 6 | Gen: 100%

#	age	bp	sg	al	su	rbc	pc	pcc	ba	bgr
1	57.0	90.0	1.01725223149487 58	0.0	0.0	0.0	0.0	0.0	0.0	116.0
2	51.0	80.0	1.01675915690774 59	0.0	0.0	0.0	0.0	0.0	0.0	123.0
3	46.0	80.0	1.01729510568522 53	0.0	0.0	0.0	0.0	0.0	0.0	146.0
4	70.0	70.0	1.01827431762569 65	0.0	0.0	0.0	0.0	0.0	0.0	102.0
5	64.0	70.0	1.01668536430068 97	0.0	0.0	0.0	0.0	0.0	0.0	138.0
6	67.0	70.0	1.01940935396313 66	0.0	0.0	0.0	0.0	0.0	0.0	113.0
7	8.0	60.0	1.025	3.0	0.0	0.0	0.0	0.0	0.0	78.0
8	60.0	70.0	1.01	1.0	0.0	nan	0.0	0.0	0.0	109.0
9	nan	70.0	1.01	3.0	0.0	0.0	0.0	0.0	0.0	110.0
10	62.0	90.0	1.02	2.0	1.0	nan	0.0	0.0	0.0	169.0

B.7 Audit Trace: Fold 7 | Gen: 100%

#	age	bp	sg	al	su	rbc	pc	pcc	ba	bgr
1	19.0	80.0	1.0196941874428282	0.0	0.0	0.0	0.0	0.0	0.0	95.0
2	52.0	70.0	1.0203584732758535	0.0	0.0	0.0	0.0	0.0	0.0	111.0
3	50.0	80.0	1.0176001193061002	0.0	0.0	0.0	0.0	0.0	0.0	117.0
4	79.0	70.0	1.0113279244554574	1.0	0.0	0.0	0.0	0.0	0.0	125.0
5	49.0	70.0	1.0158050643674237	0.0	0.0	0.0	0.0	0.0	0.0	124.0
6	65.0	70.0	1.0190405349163576	0.0	0.0	0.0	0.0	0.0	0.0	85.0
7	nan	70.0	1.02	0.0	0.0	nan	nan	0.0	0.0	219.0
8	29.0	70.0	1.02	0.0	0.0	0.0	0.0	0.0	0.0	127.0
9	50.0	90.0	1.015	1.0	0.0	1.0	1.0	0.0	0.0	nan
10	65.0	70.0	1.02	1.0	0.0	1.0	1.0	0.0	0.0	139.0

B.8 Audit Trace: Fold 8 | Gen: 100%

#	age	bp	sg	al	su	rbc	pc	pcc	ba	bgr
1	34.0	80.0	1.0189671251807937	0.0	0.0	0.0	0.0	0.0	0.0	118.0
2	59.0	80.0	1.0192939409066655	0.0	0.0	0.0	0.0	0.0	0.0	96.0
3	58.0	70.0	1.0166136679889821	0.0	0.0	0.0	0.0	0.0	0.0	130.0
4	16.0	80.0	1.0190843197981255	0.0	0.0	0.0	0.0	0.0	0.0	97.0
5	80.0	80.0	1.0226923856187611	0.0	0.0	0.0	0.0	0.0	0.0	93.0
6	48.0	80.0	1.0170508219646452	0.0	0.0	0.0	0.0	0.0	0.0	106.0
7	53.0	100.0	1.01	1.0	3.0	1.0	0.0	0.0	0.0	213.0
8	34.0	70.0	1.025	0.0	0.0	0.0	0.0	0.0	0.0	nan
9	45.0	80.0	1.02	3.0	0.0	0.0	1.0	0.0	0.0	425.0
10	33.0	80.0	1.025	0.0	0.0	0.0	0.0	0.0	0.0	89.0

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B.9 Audit Trace: Fold 9 | Gen: 100%

#	age	bp	sg	al	su	rbc	pc	pcc	ba	bgr
1	63.0	70.0	1.0158356734155976	0.0	0.0	0.0	0.0	0.0	0.0	101.0
2	63.0	80.0	1.0239045165960516	0.0	0.0	0.0	0.0	0.0	0.0	139.0
3	45.0	80.0	1.0170181692071001	0.0	0.0	0.0	0.0	0.0	0.0	104.0
4	54.0	70.0	1.0190730213667996	0.0	0.0	0.0	0.0	0.0	0.0	121.0
5	65.0	70.0	1.0173752379730303	0.0	0.0	0.0	0.0	0.0	0.0	98.0
6	66.0	70.0	1.0176539678174803	0.0	0.0	0.0	0.0	0.0	0.0	104.0
7	46.0	60.0	1.02	0.0	0.0	0.0	0.0	0.0	0.0	102.0
8	22.0	60.0	1.025	0.0	0.0	0.0	0.0	0.0	0.0	97.0
9	65.0	80.0	1.01	3.0	3.0	nan	0.0	0.0	0.0	294.0
10	23.0	60.0	1.02	0.0	0.0	0.0	0.0	0.0	0.0	95.0

B.10 Audit Trace: Fold 10 | Gen: 100%

#	age	bp	sg	al	su	rbc	pc	pcc	ba	bgr
1	61.0	70.0	1.0135738492621489	0.0	0.0	0.0	0.0	0.0	0.0	116.0
2	58.0	70.0	1.020798879114960	0.0	0.0	0.0	0.0	0.0	0.0	100.0
3	67.0	70.0	1.0187624334564942	0.0	0.0	0.0	0.0	0.0	0.0	106.0
4	44.0	80.0	1.0178275368024972	0.0	0.0	0.0	0.0	0.0	0.0	107.0
5	67.0	70.0	1.0171443150519208	0.0	0.0	0.0	0.0	0.0	0.0	95.0
6	65.0	70.0	1.0203600321727773	0.0	0.0	0.0	0.0	0.0	0.0	113.0
7	11.0	80.0	1.01	3.0	0.0	nan	0.0	0.0	0.0	nan
8	61.0	90.0	1.015	0.0	2.0	nan	0.0	0.0	0.0	nan
9	51.0	80.0	1.01	2.0	0.0	0.0	0.0	0.0	0.0	106.0
10	70.0	60.0	nan	nan	nan	nan	nan	0.0	0.0	120.0